

SCIENCE AND TECHNOLOGY OF MATERIALS – TECHNOLOGY OF METALLIC MATERIALS



CORROSION – EXERCISES

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PROBLEM 1



Two materials, gold and aluminum, are often used in engineering systems for their excellent corrosion resistance. The two sit at opposite ends of the galvanic series.

- a) Explain how both can be highly resistant to corrosion.
- b) Would it be a good idea to gold-plate aluminum for extra corrosion resistance?

PROBLEM 1 - SOLUTION



a) Gold resists corrosion because it is so resistant to oxidation. Aluminum oxidizes easily. It resists corrosion because it forms an exceptionally stable, coherent oxide that prevents further corrosion. Effectively, it paints itself.

b) Probably not. If the gold coating were penetrated the result would be a small anodic area of exposed aluminum in contact with a very large cathodic area of gold. Any damage to the oxide coating on the aluminum would cause a very rapid local corrosion.

PROBLEM 2



For reasons unspecified two segments of copper pipe are joined with a zinc couple and placed in sea water.

- a) The system corrodes through and leaks very quickly. Why? Which corrodes, the coupling or the pipe?
- b) Suggest three different ways to protect the system from corrosion, and explain why each will work.

PROBLEM 2 - SOLUTION



a) Zinc is anodic to copper in sea water. The copper and the coupling have a metallic connection through their contact and a completed circuit through the sea water. The zinc will therefore be corroded by the copper.

b) To protect the couple from corrosion we may break the electrical circuit that permits corrosion to occur or provide an alternate circuit that corrodes preferentially.

Possible solutions include the following.

(1) Paint the system on the seawater side with a paint that acts as an electrical insulator. The paint breaks the circuit by preventing direct contact between the metal and the seawater.

(2) Place an insulating sleeve between the pipe and the coupling. The sleeve breaks the metallic contact between the two materials and prevents galvanic corrosion.

(3) Note that magnesium is anodic to zinc in seawater. Place a magnesium strip across the coupling on the sea water side. The Mg is part of a circuit with the Cu and Zn and, since it is the most anodic component, corrodes preferentially.

QUESTIONS?

THE END

